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CAPSTONE PROJECT • PYTHON + SQL + POWER BI

# Beyond the Basket

*Decoding Customer Shopping Behavior Through Data*

*Turning 3,900 transactions into a clear picture of who shops, what they buy, and what drives them to spend more.*

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Presented by

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## WHAT WE'LL COVER

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# Problem Statement

In today's competitive retail landscape, businesses often lack a clear, data-backed understanding of who their customers are and what actually drives their purchasing decisions. Without that clarity, marketing spend, discount policy and subscription strategy end up built on guesswork rather than evidence.

*This capstone closes that gap by analyzing 3,900 real shopping transactions to identify exactly which customer segments, products, channels and policies drive revenue — and where the business is leaving value on the table.*

# Objectives

1

**Clean & prepare the data**

Turn raw, messy transactional data into a reliable, analysis-ready dataset.

2

**Answer real business questions**

Use SQL to answer 10 concrete questions the business actually needs answered.

3

**Quantify what drives revenue**

Measure customer segments, product performance, channels and discount effectiveness.

4

**Build a self-service dashboard**

Deliver an interactive Power BI report the business can explore on its own.

5

**Turn insight into action**

Translate every finding into one specific, actionable business decision.

# Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories — uncovering spending patterns, customer segments, product preferences and subscription behavior to guide strategic business decisions.

**3,900**

Transactions analyzed  
across the full customer base

**18**

Data columns  
spanning demographics, purchases & behavior

**3**

Analytics tools used  
Python, PostgreSQL (SQL) and Power BI

# Dataset Summary

## KEY FEATURE GROUPS

- **Customer demographics**

Age, Gender, Location, Subscription Status

- **Purchase details**

Item Purchased, Category, Amount, Season, Size, Color

- **Shopping behavior**

Discount Applied, Previous Purchases, Frequency, Rating, Shipping

## DATASET SIZE

**3,900 rows × 18 columns**

*One row per purchase transaction, covering the full customer and product footprint.*

## DATA QUALITY

**37 missing values**

Found in the Review Rating column — imputed using the median rating of each product category before analysis.

# Tools & Technologies

## Python

PANDAS

Data loading, cleaning, missing-value imputation and feature engineering.

## PostgreSQL

SQL

Structured querying to answer all 10 business questions against the cleaned data.

## Power BI

DASHBOARDING

Interactive, filterable dashboard for ongoing self-service exploration.

## CSV

RAW DATA

Source format for the 3,900-row transactional dataset before cleaning.

# Methodology

1

## Data Collection

Sourced the 3,900-row shopping transaction dataset covering demographics, purchases and behavior.

3

## Feature Engineering

Created `age_group` and `purchase_frequency_days` to enable richer segment-level analysis.

5

## SQL Analysis

Answered 10 business questions using `GROUP BY`, `CASE WHEN`, subqueries and window functions.

7

## Recommendation

Translated each SQL result into one specific, actionable business decision.

2

## Data Cleaning (Python)

Handled missing values, standardized columns to `snake_case`, and checked for data consistency.

4

## Database Integration

Loaded the cleaned DataFrame into PostgreSQL for structured, repeatable SQL analysis.

6

## Dashboarding

Visualized every finding in an interactive Power BI report with live filters.

# Data Preparation in Python

1

## Load & explore

Imported with pandas; used `df.info()` and `.describe()` to understand structure and summary statistics.

3

## Standardize columns

Renamed all columns to snake\_case for readability and consistent documentation.

5

## Check consistency

Verified `discount_applied` and `promo_code_used` were redundant, and dropped `promo_code_used`.

2

## Handle missing data

Imputed the 37 missing Review Rating values using the median rating of each product category.

4

## Engineer features

Created `age_group` (binned ages) and `purchase_frequency_days` from raw purchase data.

6

## Load to PostgreSQL

Connected Python to PostgreSQL and loaded the cleaned DataFrame for structured SQL analysis.

DEEP DIVE

# Business Questions, Queries & Insights

*10 real business questions, the SQL logic behind each answer, the exact output, and the decision it unlocks.*

## QUESTION 01

# Revenue by Gender — who generates more total spend?

C

### CONCEPT

GROUP BY aggregation with SUM

P

### PURPOSE

Compares total revenue contribution between genders to guide gender-specific marketing and inventory spend.



SQL QUERY

```
SELECT gender,  
       SUM(purchase_amount) AS revenue  
FROM shopping_data  
GROUP BY gender  
ORDER BY revenue DESC;
```

| gender | revenue |
|--------|---------|
| Male   | 157890  |
| Female | 75191   |

2 rows — one per gender, ordered by revenue descending.

# Insights & Business Decision

I

## KEY INSIGHTS



Male customers generated \$157,890 in total revenue



Female customers generated \$75,191 — less than half of male revenue

D

## BUSINESS DECISION

Investigate why male spend is over 2x female spend — it may reflect category mix or basket size. Test female-focused promotions and assortment to grow that half of the customer base.

## QUESTION 02

# High-spending customers who still use discounts

C

### CONCEPT

Scalar subquery combined with a WHERE filter on two conditions

P

### PURPOSE

Identifies customers who use discounts but are not purely price-driven, since they still spend above the average purchase amount.



SQL QUERY

```
SELECT customer_id, purchase_amount
FROM shopping_data
WHERE discount_applied = 'Yes'
AND purchase_amount > (
    SELECT AVG(purchase_amount) FROM shopping_data
)
ORDER BY customer_id;
```

| customer_id | purchase_amount |
|-------------|-----------------|
| 2           | 64              |
| 3           | 73              |
| 4           | 90              |
| 7           | 85              |
| 9           | 97              |
| 12          | 68              |
| 13          | 72              |
| 16          | 81              |

839 rows returned in total — showing the first 8. These customers discount AND spend above the \$59.76 average.

# Insights & Business Decision

I

## KEY INSIGHTS

- 839 customers used a discount while still spending above the average purchase amount
- This segment isn't purely price-driven — discounts aren't the only reason they buy

D

## BUSINESS DECISION

Target this 839-customer segment with loyalty perks and early access instead of blanket discounts — they respond to value beyond price, so margin-friendly incentives should work just as well.

### QUESTION 03

## Top 5 products by average review rating

C

### CONCEPT

GROUP BY aggregation with AVG, ranked using ORDER BY + LIMIT (Top-N)

P

### PURPOSE

Surfaces the best-reviewed products so they can be prioritised in marketing and kept well-stocked.



SQL QUERY

```
SELECT item_purchased,  
       ROUND(AVG(review_rating),2) AS average_product_rating  
FROM shopping_data  
GROUP BY item_purchased  
ORDER BY average_product_rating DESC  
LIMIT 5;
```

| item_purchased | average_product_rating |
|----------------|------------------------|
| Gloves         | 3.86                   |
| Sandals        | 3.84                   |
| Boots          | 3.82                   |
| Hat            | 3.80                   |
| Skirt          | 3.78                   |

5 rows — the highest-rated products, ordered by rating descending.

# Insights & Business Decision

I

## KEY INSIGHTS

- Gloves lead with a 3.86 average rating, closely followed by Sandals (3.84) and Boots (3.82)
- All top 5 products sit in a tight 3.78–3.86 rating band — customer satisfaction is consistently high at the top

D

## BUSINESS DECISION

Feature Gloves, Sandals, Boots, Hat and Skirt prominently in campaigns and homepage placement — proven customer satisfaction makes them safe bets for higher conversion.

#### QUESTION 04

## Shipping type comparison — Standard vs. Express

C

### CONCEPT

Filtered GROUP BY aggregation with AVG

P

### PURPOSE

Tests whether choosing faster (Express) shipping correlates with a larger average purchase amount.



SQL QUERY

```
SELECT shipping_type,  
       ROUND(AVG(purchase_amount),2) AS avg_purchase_amount  
FROM shopping_data  
WHERE shipping_type IN ('Standard','Express')  
GROUP BY shipping_type;
```

| shipping_type | avg_purchase_amount |
|---------------|---------------------|
| Standard      | 58.46               |
| Express       | 60.48               |

2 rows — one per shipping type.

# Insights & Business Decision

I

## KEY INSIGHTS

- Express shipping customers spend \$60.48 on average, versus \$58.46 for Standard
- The gap is modest (~3.5%) but consistent — Express users skew slightly higher-value

D

## BUSINESS DECISION

Promote Express shipping as a checkout upsell — it's associated with higher basket size, and even a small uplift compounds across thousands of orders.

## Subscribers vs. Non-Subscribers — spend and revenue

C

### CONCEPT

GROUP BY aggregation combining COUNT, AVG and SUM in one query

P

### PURPOSE

Quantifies the value gap between subscribers and non-subscribers to justify (or challenge) investment in the subscription program.



SQL QUERY

```
SELECT subscription_status,  
       COUNT(customer_id) AS total_customers,  
       ROUND(AVG(purchase_amount),2) AS avg_spend,  
       ROUND(SUM(purchase_amount),2) AS total_revenue  
FROM shopping_data  
GROUP BY subscription_status  
ORDER BY total_customers DESC;
```

# Query Output

| subscription_status | total_customers | avg_spend | total_revenue |
|---------------------|-----------------|-----------|---------------|
| No                  | 2847            | 59.87     | 170436.00     |
| Yes                 | 1053            | 59.49     | 62645.00      |

2 rows — subscribers vs. non-subscribers, with customer count, average spend and total revenue.

# Insights & Business Decision

I

## KEY INSIGHTS

- Non-subscribers (2,847 customers) generate \$170,436 total revenue vs. \$62,645 from 1,053 subscribers
- Average spend per customer is almost identical (\$59.87 vs \$59.49) — the gap is entirely about subscriber volume, not behavior

D

## BUSINESS DECISION

Boost subscriptions — since per-customer spend is nearly equal, converting even 10–15% of the 2,847 non-subscribers into subscribers would add meaningful recurring revenue without changing spending habits.

## Discount-dependent products

C

### CONCEPT

Conditional aggregation with CASE WHEN to compute a percentage rate

P

### PURPOSE

Flags products that rely heavily on discounting, which can signal weak full-price demand and margin risk.



SQL QUERY

```
SELECT item_purchased,  
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes'  
                           THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate  
FROM shopping_data  
GROUP BY item_purchased  
ORDER BY discount_rate DESC  
LIMIT 5;
```

| item_purchased | discount_rate |
|----------------|---------------|
| Hat            | 50.00         |
| Sneakers       | 49.66         |
| Coat           | 49.07         |
| Sweater        | 48.17         |
| Pants          | 47.37         |

5 rows — products with the highest share of discounted purchases (%).

# Insights & Business Decision

I

## KEY INSIGHTS



Hat tops the list — exactly half of all Hat purchases used a discount



Sneakers, Coat, Sweater and Pants all sit close behind, each above 47% discount usage

D

## BUSINESS DECISION

Review discount policy for Hat, Sneakers, Coat, Sweater and Pants — run a controlled test at a lower discount depth to see if full-price demand holds, protecting margin on these five items.

## Customer segmentation — New, Returning, Loyal

C

### CONCEPT

CASE WHEN bucketing on previous\_purchases to build customer segments

P

### PURPOSE

Classifies the customer base into lifecycle tiers so marketing and retention efforts can be tailored to each stage.



SQL QUERY

```
SELECT
  CASE
    WHEN previous_purchases = 0 THEN 'New'
    WHEN previous_purchases BETWEEN 1 AND 5 THEN 'Returning'
    ELSE 'Loyal'
  END AS customer_segment,
  COUNT(*) AS number_of_customers
FROM shopping_data
GROUP BY customer_segment
ORDER BY number_of_customers DESC;
```

# Query Output

| customer_segment | number_of_customers |
|------------------|---------------------|
| Loyal            | 3116                |
| Returning        | 701                 |
| New              | 83                  |

3 rows — one per lifecycle segment, ordered by size.

# Insights & Business Decision

## I KEY INSIGHTS

- Loyal customers (6+ previous purchases) make up 3,116 of the base — the vast majority
- Only 83 customers are brand New, and 701 are Returning (1–5 previous purchases)

## D BUSINESS DECISION

Launch a loyalty rewards program to retain the 3,116 Loyal customers, and build a nurture/win-back journey to move the 701 Returning customers toward Loyal status.

## QUESTION 08

# Top 3 best-selling products in each category

C

### CONCEPT

CTE + DENSE\_RANK() window function partitioned by category (Top-N per group)

P

### PURPOSE

Identifies the bestsellers within every category so merchandising and shelf space can be prioritised correctly.



SQL QUERY

```
WITH ranked_products AS (  
  SELECT category, item_purchased,  
         COUNT(*) AS total_orders,  
         DENSE_RANK() OVER (PARTITION BY category  
                             ORDER BY COUNT(*) DESC) AS item_rank  
  FROM shopping_data  
  GROUP BY category, item_purchased  
)  
SELECT item_rank, category, item_purchased, total_orders  
FROM ranked_products  
WHERE item_rank <= 3  
ORDER BY category, item_rank;
```

# Query Output

| rank | category    | item_purchased | total_orders |
|------|-------------|----------------|--------------|
| 1    | Accessories | Jewelry        | 171          |
| 2    | Accessories | Sunglasses     | 161          |
| 3    | Accessories | Belt           | 161          |
| 1    | Clothing    | Blouse         | 171          |
| 2    | Clothing    | Pants          | 171          |
| 3    | Clothing    | Shirt          | 169          |
| 1    | Footwear    | Sandals        | 160          |
| 2    | Footwear    | Shoes          | 150          |
| 3    | Footwear    | Sneakers       | 145          |
| 1    | Outerwear   | Jacket         | 163          |
| 2    | Outerwear   | Coat           | 161          |

11 rows captured — top 3 products per category (Outerwear's 3rd-ranked item was cut off in the source query result).

# Insights & Business Decision

I

## KEY INSIGHTS

- Jewelry (171 orders) and Blouse/Pants (171 each) are the single best-selling items overall
- Every category has a tight race for #1 — Accessories, Clothing and Footwear all show close order counts across their top 3

D

## BUSINESS DECISION

Prioritise inventory and marketing budget for Jewelry, Blouse, Sandals and Jacket — the #1 performer in each category — while keeping their close #2/#3 competitors well-stocked to avoid stockouts.

## QUESTION 09

# Are repeat buyers more likely to subscribe?

C

### CONCEPT

Filtered GROUP BY aggregation with COUNT

P

### PURPOSE

Tests whether customers with more than 5 previous purchases are more likely to already hold a subscription — informing upsell timing.



SQL QUERY

```
SELECT subscription_status,  
       COUNT(*) AS repeat_buyers  
FROM shopping_data  
WHERE previous_purchases > 5  
GROUP BY subscription_status  
ORDER BY repeat_buyers DESC;
```

| subscription_status | repeat_buyers |
|---------------------|---------------|
| No                  | 2518          |
| Yes                 | 958           |

2 rows — repeat buyers (>5 previous purchases) split by subscription status.

# Insights & Business Decision

I

## KEY INSIGHTS

- 2,518 repeat buyers are NOT subscribed, versus only 958 who are
- Being a repeat buyer does not predict subscription — most loyal shoppers still aren't subscribers

D

## BUSINESS DECISION

Target the 2,518 non-subscribed repeat buyers directly with subscription offers — they've already proven loyalty through repeat purchases, making them the highest-probability conversion pool.

## QUESTION 10

# Revenue contribution by age group

C

### CONCEPT

GROUP BY aggregation with SUM, using the engineered age\_group feature

P

### PURPOSE

Reveals which age segment drives the most revenue so campaigns and budget can be weighted accordingly.



SQL QUERY

```
SELECT age_group,  
       SUM(purchase_amount) AS total_revenue  
FROM shopping_data  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

# Query Output

| age_group   | total_revenue |
|-------------|---------------|
| Young Adult | 62143         |
| Middle-aged | 59197         |
| Adult       | 55978         |
| Senior      | 55763         |

4 rows — one per age group, ordered by total revenue descending.

# Insights & Business Decision

I

## KEY INSIGHTS

- Young Adults contribute the most revenue at \$62,143, narrowly ahead of Middle-aged (\$59,197)
- All four age groups are surprisingly close — the largest gap is only ~\$6,380 (about 10%)

D

## BUSINESS DECISION

Focus targeted marketing spend on Young Adult and Middle-aged segments, the top two contributors — while testing a re-engagement offer for Seniors, the lowest (but still close) contributor.

BRINGING IT TOGETHER VISUALLY

# Power BI Dashboard

An interactive dashboard was built in Power BI so the business can filter and explore these insights live — by subscription status, gender, category and shipping type.

**3.9K**

Number of Customers

**\$59.76**

Average Purchase Amount

**3.75**

Average Review Rating

**27%**

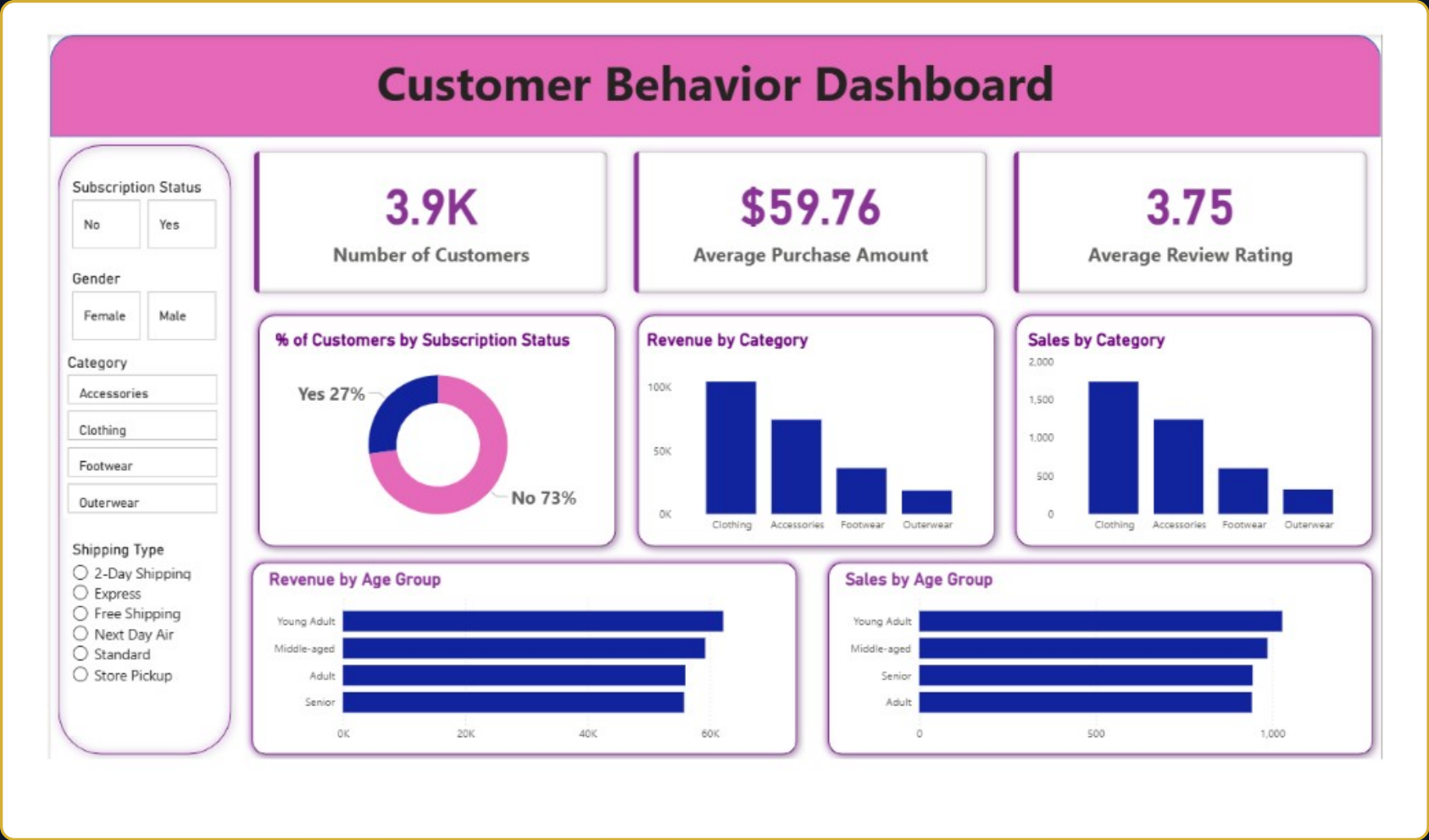
Customers Subscribed

## WHAT THE DASHBOARD LETS THE TEAM DO

- Filter live by subscription status, gender, category and shipping type
- Compare revenue and sales across product categories at a glance
- Track revenue and sales by age group side by side

# Dashboard Preview

The actual Power BI dashboard delivered as part of this project — fully interactive with live filters.



# Business Recommendations

1

## Boost Subscriptions

Promote exclusive benefits for subscribers — spend-per-customer is nearly identical to non-subscribers, so growth here is almost pure upside.

2

## Customer Loyalty Programs

Reward repeat buyers to move Returning customers into the Loyal segment, which already anchors 3,116 customers.

3

## Review Discount Policy

Balance sales boosts with margin control on the 5 most discount-dependent products.

4

## Product Positioning

Highlight top-rated and best-selling products — Gloves, Sandals, Jewelry, Blouse — in campaigns.

5

## Targeted Marketing

Focus efforts on high-revenue age groups (Young Adult, Middle-aged) and Express-shipping users.

# Conclusion

This capstone took 3,900 raw transactions through a full analytics pipeline — Python cleaning, PostgreSQL analysis and a Power BI dashboard — to answer 10 real business questions and turn each answer into a specific decision the business can act on.

**3,900**

Transactions analyzed

**10**

Business questions answered

**5**

Strategic recommendations

**1**

Interactive Power BI dashboard

*Bottom line: this project shows that a modest transactional dataset, analyzed rigorously, is enough to move a business from guesswork to evidence-based decisions.*

# Limitations & Future Scope

## LIMITATIONS

- Single-period dataset — no multi-year trend view to confirm patterns hold over time
- 37 Review Rating values were imputed, not observed — a small source of noise
- No customer-level lifetime value or churn model was built
- Location field wasn't mapped to region — geographic patterns are unexplored

## FUTURE SCOPE

- Build a predictive churn / subscription-propensity model
- Add cohort and customer-lifetime-value analysis across multiple years
- A/B test the discount-policy change recommended for the top 5 products
- Automate the dashboard with a live PostgreSQL connection for real-time refresh

WITH THANKS

# Acknowledgments

## Mentors & Reviewers

For guidance on structuring the analysis and sharpening the business recommendations.

## Open-source Tools

Python, pandas, PostgreSQL and Power BI — the stack this entire capstone was built on.

## Dataset Source

The customer shopping transaction dataset that made this end-to-end analysis possible.

**Presented by Vaishnavi Lahoti**



# Thank You

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*Questions and discussion welcome.*